Group Theory

1.

(a) The group of invertible $n \times n$ matrices over a finite field $\mathbb{F}_q$ is a finite group $\mathrm{GL}_n(\mathbb{F}_q)$. What is its order?

(b) What about the special linear group $\mathrm{SL}_n(\mathbb{F}_q)$; what is its order?

1. Solution

(a) We count the number of possibilities for each row of a matrix in $\mathrm{GL}_n(\mathbb{F}_q)$. The first row can be anything except all zeros. Since each entry has q options, there are $q^n - 1$ possibilities for the first row. The second row can be anything except a multiple of the first row. Since there are q such multiples for a given first row, there are $q^n - q$ options for the second row. Similarly, there are $q^n - q^{i-1}$ possibilities for the ith row. Therefore, $|\mathrm{GL}_n(\mathbb{F}_q)| = (q^n - 1)(q^n - q)\cdots(q^n - q^{n-1})$.

(b) Since $\mathrm{SL}_n(\mathbb{F}_q) \leq \mathrm{GL}_n(\mathbb{F}_q)$, its order must divide our answer from part (a). Suppose $A \in \mathrm{SL}_n(\mathbb{F}_q)$. Then observe that multiplying the first row of A by any nonzero element $i \in \mathbb{F}_q$ gives a matrix in $\mathrm{GL}_n(\mathbb{F}_q)$ with determinant i and every matrix of determinant i can be produced in this way. Therefore, $|\mathrm{SL}_n(\mathbb{F}_q)| = \frac{|\mathrm{GL}_n(\mathbb{F}_q)|}{q-1}$.

2. Let $1 \to A \to B \to C \to 1$ be a short exact sequence of groups. Show that if A and C are solvable, then B is also solvable.

2. Solution Since A and C are solvable, we know that each of their derived series are eventually trivial. Write these series as

$$A = A^{(0)} \geq A^{(1)} = [A, A] \geq A^{(2)} \geq \cdots \geq A^{(n)} = \{1\}$$

and

$$C = C^{(0)} \geq C^{(1)} = [C, C] \geq C^{(2)} \geq \cdots \geq C^{(m)} = \{1\}$$

where recall that $G^{(i+1)} := [G^{(i)}, G^{(i)}]$ for each group G. We want to show that the derived series for B is also eventually trivial. Under the map $B \to C$, the subgroup $B^{(m)}$ maps into $C^{(m)} = \{1\}$, so exactness implies that $B^{(m)}$ is in the image of the injection $A \to B$. Therefore, we can think of $B^{(m)} \leq A$. But then $B^{(m+n)} \leq A^{(n)} = \{1\}$, so the derived series for B stabilizes by step $m + n$. Hence, B is solvable.

3. In this problem, fix an arbitrary finite nontrivial p-group G.

(a) Show that $Z(G) \neq \{e\}$.

(b) Show that G is nilpotent.

3. Solution

(a) Let $|G| = p^n$ for some $n > 0$. The class equation gives that $|Z(G)| = p^n - \sum |G : C_G(g_i)|$ where the sum is over all non-central conjugacy classes. Since each index $|G : C_G(g_i)|$ must divide $|G|$ and be > 1, each must be p^i for some $i > 0$. Therefore, taking the class equation mod p shows that p must also divide $|Z(G)|$. Thus, $Z(G) \neq \{e\}$.

(b) We will prove this by induction on n where $|G| = p^n$. If $n = 1$, then G is cyclic, hence abelian, and is therefore nilpotent. Assume that $n > 1$ and all p-groups of order less than $|G|$ are nilpotent. By part (a), we know that $Z(G)$ is nontrivial. Since it is abelian, it is nilpotent. Furthermore, $|G/Z(G)| < |G|$ so by our induction hypothesis, $G/Z(G)$ is nilpotent. So, we have a short exact sequence $\{e\} \to Z(G) \to G \to G/Z(G) \to \{e\}$ where both of the outside groups are nilpotent. The lower central series for $Z(G)$ is $Z(G) \geq \{e\}$ and suppose for $G/Z(G)$ we have the lower central series $G/Z(G) \geq Q^1 \geq \cdots \geq Q^m = \{e\}$. Then in the lower central series for G, we see that $G^m = [G, G^{m-1}]$ maps into $Q^m = \{e\}$. Therefore, G^m is in the kernel of the map $Z(G) \to G$, i.e. $G^m \leq Z(G)$. But then $G^{m+1} = [G, G^m] = \{e\}$, so G is also nilpotent. If fact, this shows that G is of nilpotency class at most $n - 1$.

4. The additive group $\mathbb{Q}$ has the property that for any $q \in \mathbb{Q}$ and any $n \in \mathbb{Z}$, there exists an element $q' \in \mathbb{Q}$ such that $nq' = q$ (in particular $q' = q/n$). An abelian group with this property is called a *divisible* group.

(a) Show that any nontrivial divisible group is infinite.

(b) Show that if G is divisible, and N is a subgroup (normal since G is abelian), then G/N is divisible.

4. Solution

(a) We will show that if G is nontrivial and finite, then G is not divisible. If G is finite, then for any $g \in G$, $|G|g = 0$, and so if there is a nontrivial $x \in G$ there are no solutions to $|G|y = x$. Thus, G is not divisible.

(b) For any $x + N \in G/N$, we have $n(x/n + N) = x + N$, so G/N is divisible.

5. (January 2023 Problem 2) We say a group G has *big finite quotients* if for every $k > 0$, there is a normal subgroup N of G such that G/N is finite and has order at least k.

(a) Give (with proof) an example of an infinite group that does *not* have big finite quotients.

(b) Show that $\mathrm{SL}_2(\mathbb{Z})$ (the group of 2×2 integer matrices with determinant 1) has big finite quotients.

5. Solution

(a) The first infinite groups I think of are the additive groups of common rings. The group $\mathbb{Z}$ actually does have big finite quotients, so let's try $\mathbb{Q}$. As noted above, $\mathbb{Q}$ is divisible, so all of its quotients are divisible, so all of its quotients are either trivial or infinite. Hence,

$\mathbb{Q}$ does not have big finite quotients.

(b) For every $p \in \mathbb{Z}$, there is a quotient map $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ given by reducing each entry in the matrix mod p. From problem 1, $|\mathrm{SL}_2(\mathbb{F}_p)| = (p^2-1)(p^2-p)/(p-1) = p(p^2-1)$, so for any k we can certainly just choose a $p > k$ and get a big quotient.

6. (January 2016 Problem 4) Let D_k be the dihedral group of order $2k$ where $k \geq 3$.

(a) Show that the number of automorphisms of D_k is equal to $k \cdot \varphi(k)$, where φ is Euler's φ-function.

(b) Let $\mathrm{Aut}(D_k)$ denote the group of automorphisms of D_k. What is the structure of $\mathrm{Aut}(D_k)$? Describe the group as explicitly as you can.

6. Solution

(a) Recall that D_k has presentation given by

$$\langle r, f \mid r^k = f^2 = rfrf = 1 \rangle$$

where r is a rotation of the k-gon that sends each vertex to an adjacent one and f is a reflection of the k-gon through a line of symmetry. This presentation shows that every element of D_k can be written in the form $r^i f^j$, where $0 \leq i \leq k-1$ and $0 \leq j \leq 1$. Notice that the only elements of D_k of order k are those of the form r^i where i is relatively prime to k, and the only elements of order 2 are those of the form $r^i f$.

Any element of $\mathrm{Aut}(D_k)$ is determined by where it sends r and f, and r must be sent to another element of order k and f must be sent to another element of order 2.Therefore, there are $\varphi(k)$ choices for the image of r and k choices for the image of f. This shows that $|\mathrm{Aut}(D_k)| \leq k \cdot \varphi(k)$. It remains to check that each of these choices gives an automorphism of D_k. It is routine to check that each map is a homomorphism (just need to check the relations still hold), and given the map φ where $\varphi(r) = r^i$ and $\varphi(f) = r^j f$, we can construct an inverse. Since $\gcd(i,k) = 1$, there exist $a, b \in \mathbb{Z}$ such that $ai + bk = 1$, and then φ^{-1} is given by $r \mapsto r^a$ and $f \mapsto r^{-ja} f$. Thus, $|\mathrm{Aut}(D_k)| = k \cdot \varphi(k)$.

(b) By part (a), we know that any automorphism of D_k is determined by a pair of integers (j, i) where $0 \leq j \leq k-1$ and $\gcd(i,k) = 1$. Let's next understand the multiplication in this group. Given two automorphism determined by (j,i) and (n,m), we see that their composition maps $r \mapsto r^m \mapsto r^{im}$ and $f \mapsto r^n f \mapsto r^{in+j} f$. Therefore, we have $(j,i) \circ (n,m) = (in+j, im)$. This looks like it might indicate a semi-direct product. Indeed, if we consider the map $\alpha : (\mathbb{Z}/k\mathbb{Z})^* \to \mathrm{Aut}(\mathbb{Z}/k\mathbb{Z})$ which sends i to the multiplication by i map, we see that $\mathrm{Aut}(D_k) \cong \mathbb{Z}/k\mathbb{Z} \rtimes (\mathbb{Z}/k\mathbb{Z})^*$ with respect to α.

7. (August 2016 Problem 1) Let G be a group. By a maximal subgroup of G, we mean a subgroup $M \neq G$ such that the only subgroups containing M are M and G.

(a) Describe all the maximal subgroups of the dihedral group of order $2p$ where p is an odd prime. How many are there?

(b) Show that if a finite group G has only one maximal subgroup, then G is cyclic.

(c) Show that if a maximal subgroup $M \lhd G$ is normal, then the index of M in G is finite and prime.

7. Solution

(a) As in the previous problem, we have $D_p = \langle r, f \mid r^p = f^2 = rfrf = 1\rangle$ and every element of D_p can be written as $r^i f^j$ where $0 \le i \le p-1$ and $0 \le j \le 1$. Since $\varphi(p) = p-1$, every rotation element r^i with $1 \le i \le p-1$ has order p, and each of these generates the same subgroup $\langle r\rangle \cong \mathbb{Z}/p\mathbb{Z}$. Every element of the form $r^i f$ with $0 \le i \le p-1$ has order 2, and each generates its own subgroup $\langle r^i f\rangle \cong \mathbb{Z}/2\mathbb{Z}$. Since $|D_p| = 2p$, we know that any maximal subgroup must have order 2 or p. Therefore, the maximal subgroups are exactly the single subgroup $\langle r\rangle$ of order p and the p subgroups $\langle r^i f\rangle$ of order 2.

(b) Suppose $M < G$ is the only maximal subgroup of G. Then $M \neq G$, so there exists some $g \in G \setminus M$. Then $\langle g\rangle$ is a nontrivial subgroup of G, so it is either contained in a maximal subgroup of all of G (since every proper subgroup must be contained in a maximal subgroup). Since, $g \notin M$, we must have that $\langle g\rangle = G$. Thus, G is cyclic.

(c) Suppose $M \lhd G$ is maximal. Then the lattice theorem gives that the only subgroups of G/M are the trivial group and the whole group. Let $g \in G/M$ be any nontrivial element (which must exist by definition of M). Then $\langle g\rangle = G/M$, so G/M is cyclic. Observe that $G/M \ncong \mathbb{Z}$ since $\mathbb{Z}$ has proper subgroups such as $2\mathbb{Z}$. Therefore, $G/M \cong \mathbb{Z}/n\mathbb{Z}$ for some $n > 1$. Let p be any prime factor of n. Then $\langle g^{\frac{n}{p}}\rangle$ is a subgroup of G of order p. Since G/M has no proper subgroups, we must have that $n = p$, so $|G : M| = p$.

8. (January 2019 Problem 2) If G is a group and $x, y \in G$, then we let $\langle x, y\rangle$ denote the subgroup of G generated by x and y. For $x \in G$, define $\mathrm{Cyc}_G(x) := \{y \in G \mid \langle x, y\rangle \text{ is cyclic}\}$.

(a) Show by example that $\mathrm{Cyc}_G(x)$ need not be a subgroup of G.

(b) Let $\mathrm{Cyc}(G) = \bigcap_{x\in G} \mathrm{Cyc}_G(x)$. Show that $\mathrm{Cyc}(G)$ is a subgroup of G.

(c) Show that $\mathrm{Cyc}(G)$ lies in the center of G.

(d) Assume now that G is finite. Show that $\mathrm{Cyc}(G)$ is cyclic.

8. Solution

(a) Consider $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$ and $x = (0, 2)$. Then $(1, 1)$ and $(0, 1)$ are in $\mathrm{Cyc}_G(x)$ since $\langle (0, 2), (1, 1)\rangle = \langle (1, 1)\rangle)$ and $\langle (0, 2), (0, 1)\rangle = \langle (0, 1)\rangle$, but $(1, 0) = (1, 1) - (0, 1)$ is not in $\mathrm{Cyc}_G(x)$ since $\langle (0, 2), (1, 0)\rangle \cong \mathbb{Z}/2\mathbb{Z} \times 2\mathbb{Z}$.

(b) We first show that $\mathrm{Cyc}(G)$ is closed under multiplication. Let $y, z \in \mathrm{Cyc}(G)$. Then for every $x \in G$ we have $\langle x, y\rangle = \langle g_{xy}\rangle$ and $\langle x, z\rangle = \langle g_{xz}\rangle$ for some elements $g_{xy}, g_{xz} \in G$. In particular, $\langle g_{xy}, z\rangle = \langle g_{g_{xy}z}\rangle$. Then $\langle yz\rangle \le \langle y, z\rangle \le \langle g_{xy}, z\rangle = \langle g_{g_{xy}z}\rangle$, so $\langle x, yz\rangle$ must be cyclic for each $x \in G$ since it is a subgroup of the cyclic group $\langle g_{g_{xy}z}\rangle$. Thus, $yz \in \mathrm{Cyc}(G)$. Now we show that it is closed under taking inverses. Let $y \in \mathrm{Cyc}(G)$. We need to show that for each $x \in G$, $\langle x, y^{-1}\rangle$ is cyclic. But $\langle x, y^{-1}\rangle \leq \langle x, y\rangle$, which is cyclic by assumption, so $y^{-1} \in \operatorname{Cyc}(G)$.

(c) Let $y \in \operatorname{Cyc}(G)$. We want to show that $y \in Z(G)$, that is, that y commutes with every element of G. We know that for any $x \in G$, $\langle x, y\rangle$ is cyclic, and hence abelian. Therefore, x and y must commute.

(d) First, observe that $\operatorname{Cyc}(G) \leq \operatorname{Cyc}(Z(G))$ and since a subgroup of a cyclic group is cyclic, it is enough to show that $\operatorname{Cyc}(Z(G))$ is cyclic. Since $Z(G)$ is abelian, it suffices to show that $\operatorname{Cyc}(G)$ is cyclic when G is abelian. By the fundamental theorem of finitely generated abelian groups, we can write $G = \prod_{i=1}^k \mathbb{Z}/p_i^{e_i}\mathbb{Z}$ for some primes p_i and exponents $e_i \geq 1$. Let I be the set of indices i such that $p_i \neq p_j$ for any index $j \neq i$. We will show that $\operatorname{Cyc}(G) = \prod_{i\in I} \mathbb{Z}/p_i^{e_i}\mathbb{Z}$, which is a cyclic group since no p_i is repeated.

To show that $\operatorname{Cyc}(G) \subseteq \prod_{i\in I} \mathbb{Z}/p_i^{e_i}\mathbb{Z}$, suppose instead that there is a $y \in \operatorname{Cyc}(G)$ that is not in the RHS. Then, there must exist indices $i, j \in \{1, \ldots, k\} \setminus I$ such that $p_i = p_j = p$ where y has a nonzero component in at least one of the factors $\mathbb{Z}/p^{e_i}\mathbb{Z}$ or $\mathbb{Z}/p^{e_j}\mathbb{Z}$ of G. Let (a, b) be the image of y under the projection from G to $\mathbb{Z}/p^{e_i}\mathbb{Z} \times \mathbb{Z}/p^{e_j}\mathbb{Z}$, and assume without loss of generality that $a \neq 0$. Let $x \in G$ be any element that projects to $(0, 1)$. Since $\langle x, y\rangle$ is cyclic, its image must also be cyclic. But, $\langle (0, 1), (a, b)\rangle \cong \mathbb{Z}/(p^{e_i}/\gcd(a, p^{e_i}))\mathbb{Z} \times \mathbb{Z}/p^{e_j}\mathbb{Z}$, which is not cyclic since $p^{e_i}/\gcd(a, p^{e_i})$ and p^{e_j} are nonzero powers of p.

To show the reverse inclusion, let $A = \prod_{i\in I} \mathbb{Z}/p_i^{e_i}\mathbb{Z}$, and write $G = A \times B$, where B consists of all of the remaining factors for $i \neq I$. Let $(a, 0) \in A$. We want to show that $(a, 0) \in \operatorname{Cyc}(G)$, i.e. that for any $x = (x_A, x_B) \in G$ the subgroup $\langle (x_A, x_B), (a, 0)\rangle$ is cyclic. Consider that

$$\langle (x_A, x_B), (a, 0)\rangle \leq \langle (x_A, 0), (0, x_B), (a, 0)\rangle = \langle \langle (x_A, 0), (a, 0)\rangle, (0, x_B)\rangle.$$

Since $\langle (x_A, 0), (a, 0)\rangle$ is a subgroup of A, it is cyclic, so let $\langle (x_A, 0), (a, 0)\rangle = \langle a'\rangle$ and suppose its order is n. Note that n is divisible only by primes p_i with $i \in I$. Then $(0, x_B)$ also generates a cyclic subgroup of G, and its order must be coprime to n. Therefore,

$$\langle (x_A, x_B), (a, 0)\rangle \leq \langle (a', 0), (0, b)\rangle \leq \langle a'\rangle \times \langle b\rangle.$$

Notice that the RHS is cyclic since it is a product of cyclic groups of coprime order. Thus, the LHS is cyclic as a subgroup of a cyclic group. Hence, $\operatorname{Cyc}(G) = A$.

9. (January 2023 Problem 5) Let p be a prime number and let G be the group of 2×2 matrices of the form

$$\begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix}, \qquad a \in \mathbb{F}_p,\ b \in \mathbb{F}_p^\times.$$

Fix a dimension n, and let $\rho\colon G \to \mathrm{GL}_n(\mathbb{C})$ be a homomorphism. Put

$$x = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},\ y_b = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} \qquad (b \in \mathbb{F}_p^\times).$$

(a) Show that $\rho(x)$ is diagonalizable, and that its eigenvalues are pth roots of unity.

(b) Suppose $v \in \mathbb{C}^n$, $v \neq 0$ is an eigenvector of $\rho(x)$ of eigenvalue λ. Prove that for any $b \in \mathbb{F}_p^\times$, $\rho(y_b)v$ is also an eigenvector of $\rho(x)$, and find its eigenvalue.

(c) Let v be as in part (b). Show that $W = \operatorname{span}\langle \rho(y_b)v : b \in \mathbb{F}_p^\times \rangle$ is G-invariant (that is, $\rho(g)W \subseteq W$ for all $g \in G$), and that, if $\lambda \neq 1$, $\dim W = p - 1$.

9. Solution

(a) Note that in G, $x^p = I$, so $\rho(x)$ satisfies the polynomial $x^p - 1$. Thus, all the eigenvalues of $\rho(x)$ must be roots of $x^p - 1$, i.e. pth roots of unity. Furthermore, the minimal polynomial of $\rho(x)$ divides $x^p - 1$, which has all distinct roots, so the minimal polynomial must have all distinct roots, so $\rho(x)$ must be diagonalizable.

(b) In G, we have that $y_b x^b = x y_b$. Thus, $\rho(x)\rho(y_b)v = \rho(y_b)\rho(x)^b v = \lambda^b \rho(y_b)v$. Thus, $\rho(y_b)v$ is an eigenvector of eigenvalue λ^b.

(c) First note that x and the set of all y_b together generate G, since

$$\begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix} = y_b x^a.$$

Thus, since each of the generators of W is an eigenvector of $\rho(x)$, and these eigenvectors are permuted by each $\rho(y_b)$, the generating set of W is fixed by the action of G, and hence W is G-invariant. If $\lambda \neq 1$, λ is a primitive pth root of unity, and so each λ^b is distinct. Thus, the $\rho(y_b)v$ are linearly independent, since they are eigenvectors of $\rho(x)$ of different eigenvalues. Since there are $p - 1$ choices for b, this means $\dim W = p - 1$.